

Question:

At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<think>

To determine which part of the brain is least developed at birth , let's evaluate the development stages of the different brain regions mentioned .

- The brain stem , which controls vital functions like breathing , heart rate , and sleep , is usually considered fully developed at birth .

- The visual system, responsible for processing visual information, is also mostly developed at birth, as newborns can perceive and respond to visual stimuli.

- The limbic system , involved in emotions , memory , and motivation , is also largely developed at birth .

- The basal ganglia , involved in movement control and habit formation , are relatively developed at birth .

- The thalamus, which relays sensory and motor signals to the cortex, is largely developed at birth.

- The parietal lobe, responsible for processing sensory information and spatial awareness, is relatively developed at birth.

- The cerebellum, which coordinates muscle movements and balance, is largely developed at birth.

- The cortex, the outer layer of the brain responsible for higher-order functions like thought, action, and perception, is one of the last parts to develop fully, with significant maturation happening post-birth through infancy and childhood.

- The hypothalamus, which regulates body temperature, hunger, thirst, and other basic bodily functions, is relatively developed at birth.

Therefore , the least developed part of the brain at birth is the cortex , which continues to mature significantly in the first few years of life .

</think><answer> G . cortex